

numerical expression

Algebra 1 • 1.1

power

Algebra 1 • 1.1

base

Algebra 1 • 1.1

exponent

Algebra 1 • 1.1

evaluate

Algebra 1 • 1.1

order of operations

Algebra 1 • 1.1

variable

Algebra 1 • 1.2

algebraic expression

Algebra 1 • 1.2

An expression with a base and an exponent that shows repeated multiplication.

$$2^5 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 32$$

A combination of numbers and operations — no variables and no equal sign.

Example: $3(8 - 2) + 4^2$

In a power, the number that tells how many times the base is used as a factor.

In 7^3 , the exponent is 3.

In a power, the number that is used as a factor.

In 7^3 , the base is 7.

The agreed sequence for evaluating: Grouping symbols, Exponents, Multiply/Divide left to right, Add/Subtract left to right.

To find the value of an expression by carrying out all of its operations.

An expression containing at least one variable, with numbers and operations.

Example: $4x - 7$

A letter or symbol used to represent an unknown or changing value.

term

Algebra 1 • 1.2

constant

Algebra 1 • 1.2

coefficient

Algebra 1 • 1.4

define a variable

Algebra 1 • 1.2

additive identity

Algebra 1 • 1.3

multiplicative identity

Algebra 1 • 1.3

additive inverse

Algebra 1 • 1.3

multiplicative inverse (reciprocal)

Algebra 1 • 1.3

A term that is a number only — its value never changes.

In $3x^2 + 5x - 8$, the constant is -8 .

A number, a variable, or a product or quotient of numbers and variables. Terms are separated by $+$ and $-$ signs.

$3x^2 + 5x - 8$ has three terms.

To choose a letter and state exactly what it represents, including units.

Let h = the number of hours worked.

The numerical factor of a term containing a variable.

In $5x$, the coefficient is 5. In x , it is 1.

One. Multiplying any number by 1 leaves it unchanged: $a \cdot 1 = a$.

Zero. Adding 0 to any number leaves it unchanged: $a + 0 = a$.

The number you multiply by to get 1. For $a \neq 0$, the multiplicative inverse of a is $1/a$: $a \cdot (1/a) = 1$.

The reciprocal of $\frac{2}{3}$ is $\frac{3}{2}$.

The opposite of a number. A number plus its additive inverse equals 0: $a + (-a) = 0$.

Multiplicative Property of Zero

Algebra 1 • 1.3

Commutative Property

Algebra 1 • 1.3

Associative Property

Algebra 1 • 1.3

Distributive Property

Algebra 1 • 1.4

Reflexive Property

Algebra 1 • 1.3

Symmetric Property

Algebra 1 • 1.3

Transitive Property

Algebra 1 • 1.3

Substitution Property

Algebra 1 • 1.3

Order does not matter when adding or multiplying: $a + b = b + a$ and $ab = ba$.

Any number times 0 is 0: $a \cdot 0 = 0$.

Multiply the outside factor by EACH term inside the parentheses: $a(b + c) = ab + ac$.

$$3(x + 4) = 3x + 12$$

Grouping does not matter when adding or multiplying: $(a + b) + c = a + (b + c)$ and $(ab)c = a(bc)$.

An equation can be flipped: if $a = b$, then $b = a$.

Any quantity equals itself: $a = a$.

If $a = b$, then a may be replaced by b in any expression.

Equality chains together: if $a = b$ and $b = c$, then $a = c$.

like terms

Algebra 1 • 1.4

simplest form

Algebra 1 • 1.4

equivalent expressions

Algebra 1 • 1.4

absolute value

Algebra 1 • 1.5

descriptive modeling

Algebra 1 • 1.6

metric

Algebra 1 • 1.6

accuracy

Algebra 1 • 1.6

precision

Algebra 1 • 1.6

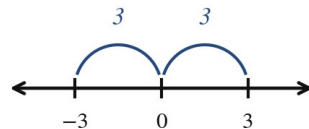
An expression with no like terms left to combine and no parentheses.

Terms with exactly the same variable(s) raised to the same power(s).

$4x$ and $-9x$ are like terms; $4x$ and $4x^2$ are not.

A number's distance from 0 on a number line — never negative.

$|-3| = 3$ and $|3| = 3$.



Expressions that have the same value for every value of the variable.

$2(x + 3)$ and $2x + 6$

A rule or unit chosen to measure something, so results can be compared.

Example: miles per gallon.

Using data and mathematics to describe a real situation — choosing what to measure and how to represent it.

How exact a measurement is — how finely it is expressed. (Precise is not the same as accurate.)

How close a measurement or model is to the true value.